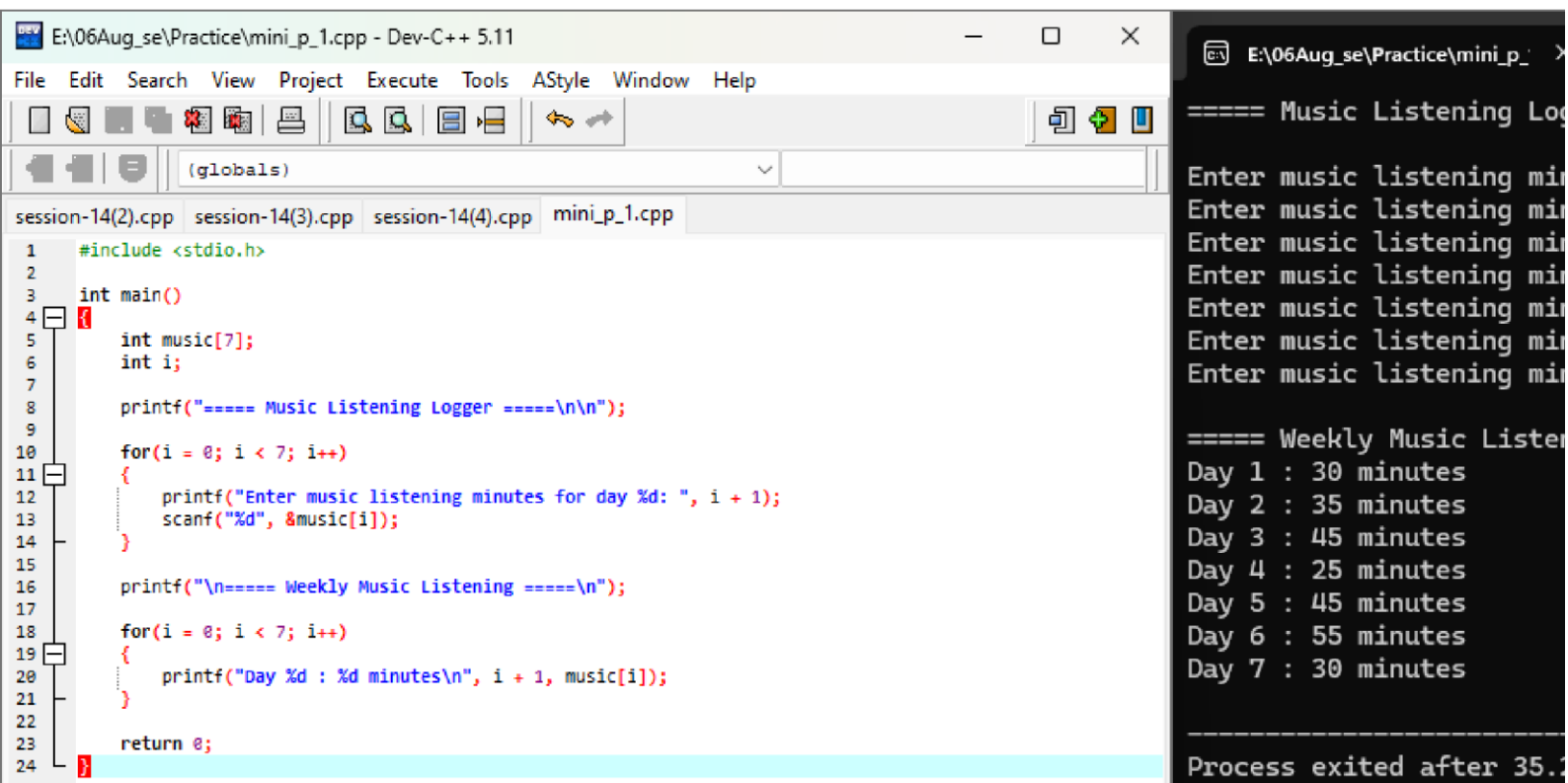


MINI PROJECT

Task : 1

Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.



The screenshot displays the Dev-C++ IDE with the file 'mini_p_1.cpp' open. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int music[7];
6     int i;
7
8     printf("==== Music Listening Logger ==== \n\n");
9
10    for(i = 0; i < 7; i++)
11    {
12        printf("Enter music listening minutes for day %d: ", i + 1);
13        scanf("%d", &music[i]);
14    }
15
16    printf("\n==== Weekly Music Listening ==== \n\n");
17
18    for(i = 0; i < 7; i++)
19    {
20        printf("Day %d : %d minutes\n", i + 1, music[i]);
21    }
22
23    return 0;
24 }
```

The console window on the right shows the program's execution. It prompts the user to enter music listening minutes for each of the 7 days. The entered values are 30, 35, 45, 25, 45, 55, and 30 minutes respectively. The program then displays the weekly summary and exits after 35.2 seconds.

```
==== Music Listening Logger
Enter music listening minutes for day 1: 30
Enter music listening minutes for day 2: 35
Enter music listening minutes for day 3: 45
Enter music listening minutes for day 4: 25
Enter music listening minutes for day 5: 45
Enter music listening minutes for day 6: 55
Enter music listening minutes for day 7: 30

==== Weekly Music Listening
Day 1 : 30 minutes
Day 2 : 35 minutes
Day 3 : 45 minutes
Day 4 : 25 minutes
Day 5 : 45 minutes
Day 6 : 55 minutes
Day 7 : 30 minutes

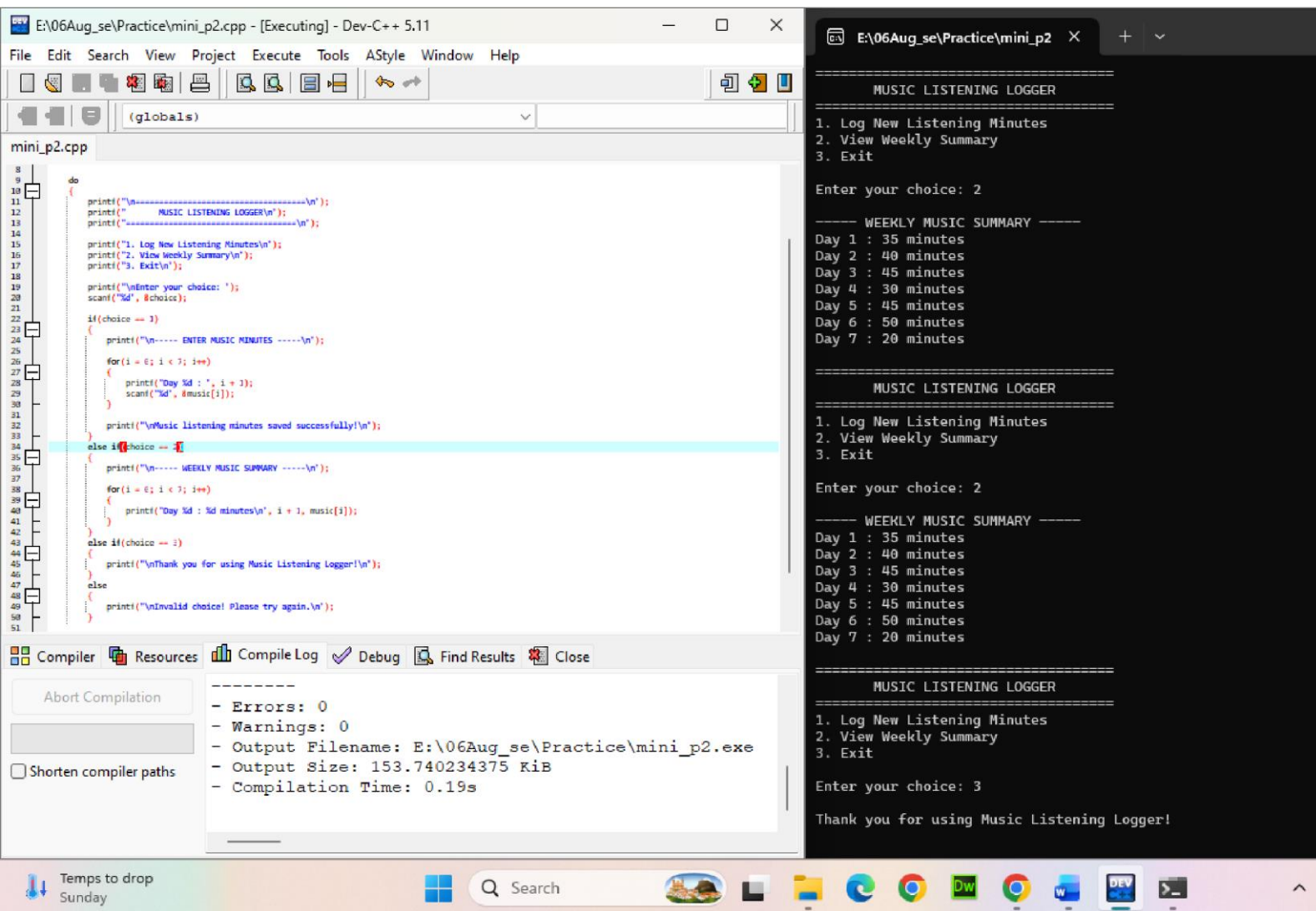
Process exited after 35.2 seconds
```

Create a Music Listening Logger program. Enter music listening minutes for Day 1 to Day 7 and store them in an array. Then display the minutes for each day.

Task : 2

Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

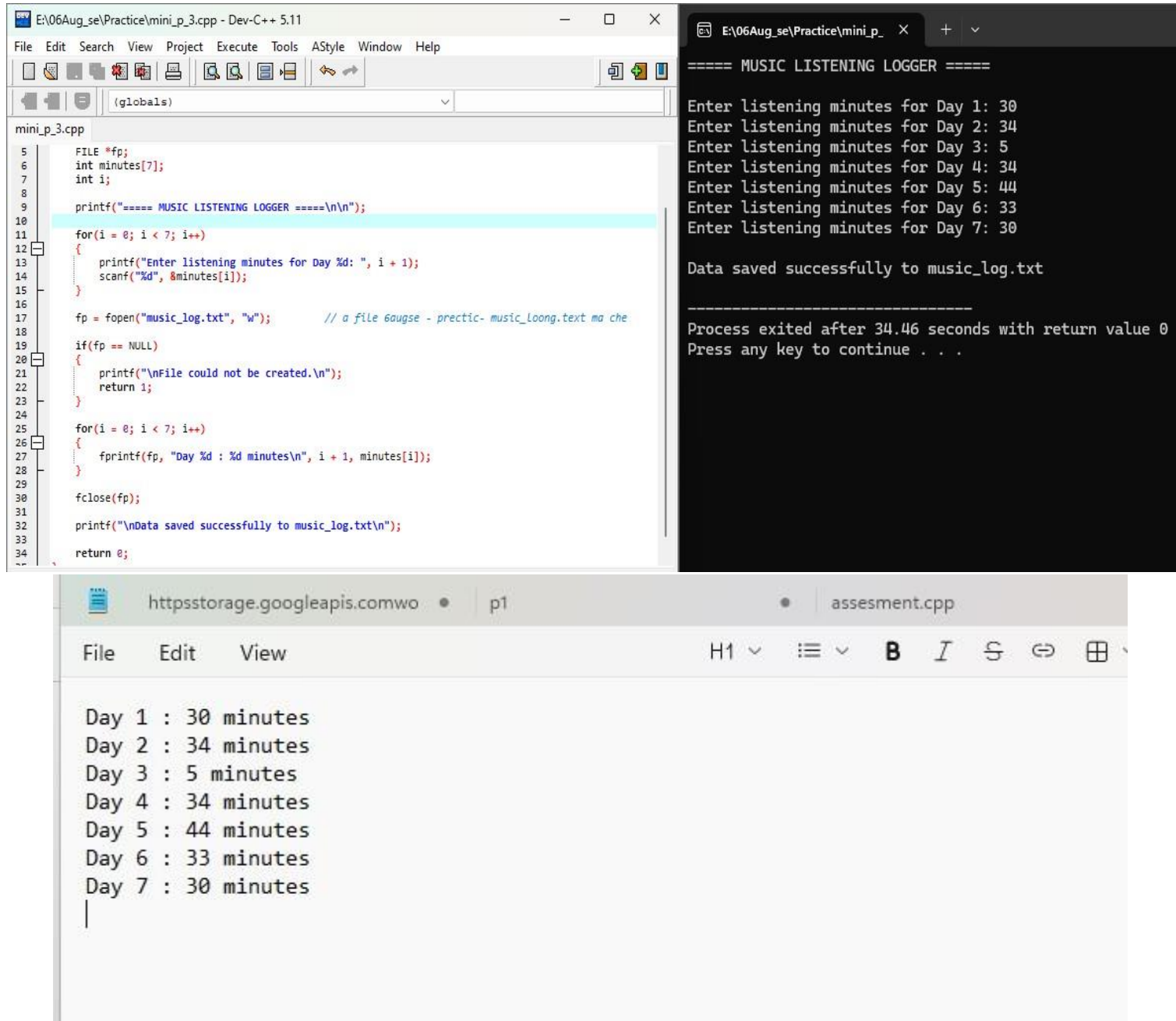
Hint: Use a loop to repeatedly show the menu until the user selects exit.



Task : 3

Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named `music_log.txt` for persistence.

Ans.



The image shows a C++ program in Dev-C++ and its execution output. The program, `mini_p_3.cpp`, prompts the user to enter listening minutes for 7 days and saves the data to `music_log.txt`. The execution output shows the program running successfully and displaying the saved data.

```
5 FILE *fp;
6 int minutes[7];
7 int i;
8
9 printf("==== MUSIC LISTENING LOGGER ====\n");
10
11 for(i = 0; i < 7; i++)
12 {
13     printf("Enter listening minutes for Day %d: ", i + 1);
14     scanf("%d", &minutes[i]);
15 }
16
17 fp = fopen("music_log.txt", "w"); // a file 6augse - prectic- music_logong.txt ma che
18
19 if(fp == NULL)
20 {
21     printf("\nFile could not be created.\n");
22     return i;
23 }
24
25 for(i = 0; i < 7; i++)
26 {
27     fprintf(fp, "Day %d : %d minutes\n", i + 1, minutes[i]);
28 }
29
30 fclose(fp);
31 printf("\nData saved successfully to music_log.txt\n");
32
33 return 0;
```

```
==== MUSIC LISTENING LOGGER ====
Enter listening minutes for Day 1: 30
Enter listening minutes for Day 2: 34
Enter listening minutes for Day 3: 5
Enter listening minutes for Day 4: 34
Enter listening minutes for Day 5: 44
Enter listening minutes for Day 6: 33
Enter listening minutes for Day 7: 30

Data saved successfully to music_log.txt

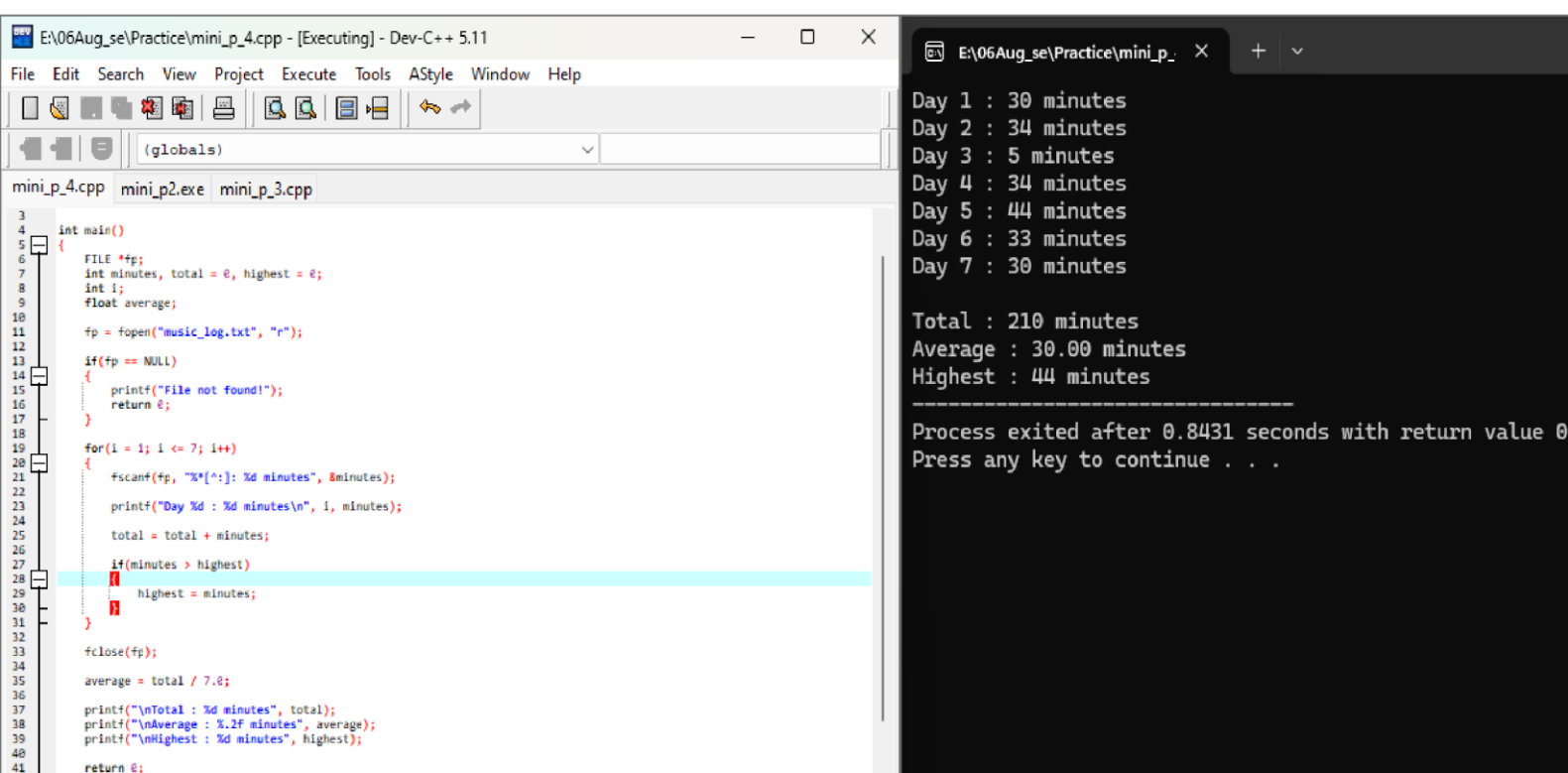
-----
Process exited after 34.46 seconds with return value 0
Press any key to continue . . .
```

```
Day 1 : 30 minutes
Day 2 : 34 minutes
Day 3 : 5 minutes
Day 4 : 34 minutes
Day 5 : 44 minutes
Day 6 : 33 minutes
Day 7 : 30 minutes
|
```

File 6augse – practice – music_log.txt

Task-4

Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest



The screenshot displays a C++ IDE with the file `mini_p_4.cpp` open. The code reads a file named `music_log.txt` and calculates the total, average, and highest listening minutes for a week. The execution output shows the following results:

```
Day 1 : 30 minutes
Day 2 : 34 minutes
Day 3 : 5 minutes
Day 4 : 34 minutes
Day 5 : 44 minutes
Day 6 : 33 minutes
Day 7 : 30 minutes

Total : 210 minutes
Average : 30.00 minutes
Highest : 44 minutes

-----
Process exited after 0.8431 seconds with return value 0
Press any key to continue . . .
```

listening minutes in the week.

Ans.

Day 1 : 30 minutes

Day 2: 34 minutes

Day 3: 5 minutes

Day 4: 34 minutes

Day 5: 44 minutes

Day 6: 33 minutes

Day 7 : 30 minutes

Total : 210 minutes

Average : 30.00 minutes

Highest : 44 minutes

Task-5

Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.

Ans.

The screenshot shows a C++ IDE with the following components:

- Main Window:** Displays the source code for `mini_p_5.cpp`. The code implements a music listening logger with functions for logging, displaying data, and resetting the data.
- Bottom Panel:** Shows the compiler output, indicating successful compilation with 0 errors and 0 warnings. The output includes the filename `E:\06Aug_se\Practice\mini_p_5.exe`, the output size `154.751953125 KiB`, and the compilation time `0.19s`.
- Right Panel:** Shows the program's output, displaying the listening minutes for 7 days and the option to reset the data.

=====

MUSIC LISTENING LOGGER

=====

- Log New Listening Minutes
- View Weekly Summary
- Reset Weekly Data
- Exit

Enter your choice: 1

Enter listening minutes for 7 days:

Day 1: 30

Day 2: 35

Day 3: 40

Day 4: 45

Day 5: 55

Day 6: 50

Day 7: 30

Data saved successfully!

1. Log New Listening Minutes

=====

MUSIC LISTENING LOGGER

=====

2. View Weekly Summary
3. Reset Weekly Data
4. Exit

Enter your choice: 2

===== WEEKLY SUMMARY =====

Day 1 : 30 minutes

Day 2 : 35 minutes

Day 3 : 40 minutes

Day 4 : 45 minutes

Day 5 : 55 minutes

Day 6 : 50 minutes

Day 7 : 30 minutes

Total : 285 minutes

Average : 40.71 minutes

Highest : 55 minutes

1. Log New Listening Minutes

=====

MUSIC LISTENING LOGGER

=====

- 2. View Weekly Summary
- 3. Reset Weekly Data
- 4. Exit

Enter your choice: 3

Are you sure you want to reset? (y/n): n

Reset cancelled!

=====

MUSIC LISTENING LOGGER

=====

- 1. Log New Listening Minutes
- 2. View Weekly Summary
- 3. Reset Weekly Data
- 4. Exit

Enter your choice: 4

Thank you for using Music Listening Logger!

